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Project Report

DESIGN AND IMPLEMENTATION OF A 4×4 REGISTER FILE

with Synchronous Read/Write and Asynchronous Clear

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Abstract

This report presents the design, construction, and verification of a 4×4 register file implemented using standard 74-series TTL integrated circuits. The system provides four 4-bit storage registers, each addressable via a 2-bit address bus, and supports three fundamental operations: synchronous clock-triggered write, bus-buffered read with 3-state output, and asynchronous active-low clear. The design employs a 74139 dual 2-to-4 address decoder, four 74175 quad D flip-flop ICs as the storage array, two 74LS244 octal bus buffers for tri-state read control, a 7402 NOR gate array for clock gating, a 7408 AND gate array for write-enable logic, and a 7432 OR gate array for selective clear functionality. The complete system consumes nine ICs, stores 16 bits of data, and operates from a single +5 V supply. This project demonstrates core digital design principles including register-file architecture, address decoding, bus multiplexing, and synchronous/asynchronous mixed-mode control—foundational concepts in computer organisation and digital systems design.

Keywords: Register file, 74175 flip-flop, 74139 decoder, 74LS244 tri-state buffer, TTL logic, synchronous write, asynchronous clear, bus architecture.

1. Introduction

A register file is a small, fast memory array within a processor or digital system that holds operands for ongoing computations. It forms the topmost tier of the memory hierarchy, offering the lowest latency at the cost of limited capacity. Every modern microprocessor—from simple 8-bit controllers to 64-bit CPUs—incorporates a register file as the primary working storage for the arithmetic-logic unit (ALU).

This project realises a minimal but functionally complete register file at the discrete-logic level. By constructing the file from individual 74-series ICs rather than relying on a pre-integrated memory device, the designer gains direct insight into the fundamental mechanisms of address decoding, clocked storage, tri-state bus control, and asynchronous reset—mechanisms that underpin every stored-program computer.

1.1 Objectives

- Design a 4×4 register file (4 rows × 4 bits per row) using standard TTL ICs.
- Implement 2-bit address decoding to select one of four registers.
- Provide synchronous write, asynchronous clear, and tri-state read operations.
- Verify correct operation through systematic functional testing.
- Document the design to professional engineering standards.

1.2 Scope

The project is limited to a combinational-and-sequential logic implementation using 74-series TTL devices. FPGA, HDL, or microcontroller implementations are outside scope. The design targets a single-clock, single-port read/write architecture; simultaneous read and write to the same address is not supported (and would cause a bus conflict) and is therefore excluded from valid operating modes.

2. Background Theory

2.1 Register File Architecture

A register file consists of an array of registers, an address decoder, write-enable logic, and a read bus. During a write cycle, the address decoder activates one register's clock line; the data present on the data-input bus is latched on the rising edge of the clock. During a read cycle, the decoder enables the tri-state output buffers of the selected register, driving its contents onto the shared bus, while all other registers' outputs remain in the high-impedance (High-Z) state to prevent bus contention.

2.2 D Flip-Flop (74175)

The 74175 is a quad positive-edge-triggered D flip-flop with a common clock and an active-low asynchronous clear. On the rising edge of the clock (CLK), the logic level at each D input is transferred to the corresponding Q output and held until the next active clock edge. The asynchronous clear (CLR\) input overrides the clock: when CLR\ is driven LOW, all four Q outputs are immediately set to logic 0, independent of clock state. This mixed synchronous/asynchronous behaviour is essential for the register file's write and clear operations.

2.3 2-to-4 Decoder (74139)

The 74139 contains two independent 2-to-4 binary decoders. Each decoder accepts two address inputs (A0, A1) and an active-low enable (G\). When enabled, exactly one of the four active-low outputs (Y0–Y3) is driven LOW according to the binary value of the address inputs; all other outputs remain HIGH. The active-low nature of the outputs is directly compatible with the active-low clear input of the 74175 and the active-low enable of the 74LS244.

2.4 Tri-State Bus Buffer (74LS244)

The 74LS244 is an octal (8-channel) non-inverting buffer/line driver with two groups of four buffers, each group controlled by an independent active-low output-enable (G\). When G\ is LOW, the buffer drives its output to match the input logic level. When G\ is HIGH, the output is disconnected (High-Z), effectively removing that device from the bus. This tri-state property enables multiple register outputs to share a single bus without contention, provided at most one enable is asserted at any time.

2.5 Write-Enable Gating Logic

Write gating is achieved using two standard logic functions. First, a NOR gate (7402) combines the active-low row-select signal from the decoder with the clock input. The NOR output is HIGH only when both inputs are LOW—i.e., when the row is selected (select = 0) and the clock is in its LOW phase. This signal is then ANDed (7408) with the active-high Write Enable control signal. The AND gate output forms the qualified clock pulse delivered to the 74175 CLK pin of the selected row, ensuring that data is stored only in the addressed register and only when the Write Enable is asserted.

2.6 Clear Logic

Clear control is implemented with a 7432 OR gate per row. One input of each OR gate is connected to the global Clear button (active LOW when pressed), and the other input is connected to the row-select output from the decoder. When the Clear button is pressed (LOW), the OR output goes LOW regardless of the row-select state, asserting CLR\ on all 74175 ICs simultaneously—a global clear. When the button is not pressed (HIGH), the OR output mirrors the row-select signal, meaning CLR\ is always HIGH (inactive) for rows that are not selected, preserving their contents during normal operation.

3. System Design

3.1 Top-Level Architecture

The system is partitioned into five functional subsystems: (1) Address Decoding, (2) Register Array, (3) Write Logic, (4) Read Bus, and (5) Clear Logic. External control signals are a 2-bit address (A0, A1), a single-bit Write Enable, a single-bit Read Enable (used as the decoder's G\ input), a Clock signal, and an active-low Clear button. Data enters and leaves the system as a 4-bit bus (D0–D3 for input; LED-driven bus for output).

3.2 IC Inventory

Qty	Part No.	Type	Role in System
4	74175	Quad D Flip-Flop	4-bit registers (one per row)
2	74LS244	Octal 3-State Buffer	Read bus multiplexer
1	74139	Dual 2-to-4 Decoder	Address decoding
1	7402	Quad 2-Input NOR	Clock gating
1	7408	Quad 2-Input AND	Write-enable logic
1	7432	Quad 2-Input OR	Clear logic
4	LED	Light Emitting Diode	Data output display
4	220 Ω	Resistor	LED current limiting
5	0.1 µF	Ceramic Capacitor	Power supply decoupling

3.3 Address Decoding Subsystem

The first decoder section of the 74139 is used. The Read Enable signal drives the active-low enable pin (1G\): when Read Enable is LOW, the decoder is active and its outputs faithfully reflect the address. Address lines A0 and A1 connect to the 1A and 1B inputs respectively. The four active-low outputs (1Y0–1Y3) drive the row-select signals for Rows 0–3. These signals simultaneously gate the read bus buffers and feed the clear logic; the write logic derives its clock-gate signals from the same row-select lines through the NOR–AND chain described in Section 2.5.

3.4 Register Array

Four 74175 ICs (IC2–IC5) serve as the register array. Each IC stores one 4-bit row. All four D inputs (D0–D3) are connected in parallel across all rows; data written depends solely on which row's clock is activated. Q outputs feed the 74LS244 read buffers. The CLR\ inputs are driven by the clear logic OR gates. Each 74175 VCC pin has an individual 0.1 µF ceramic bypass capacitor to prevent switching noise from propagating through the power rail.

3.5 Write Logic

The 7402 NOR IC provides one NOR gate per row (gates 2–4 are used; gate 1 is unused). For Row n, the NOR inputs are the row-select line (active LOW when selected) and the system Clock. The NOR output is HIGH only when row n is selected AND the clock is LOW. This is equivalent to a gated, phase-inverted clock. The 7408 AND IC's corresponding gate combines the NOR output with the Write Enable signal; the AND output drives the 74175 CLK pin. Because the 74175 is positive-edge-triggered, data is latched on the 0→1 transition of the AND output, which corresponds to the rising edge of the system clock—provided the row is addressed and Write Enable is HIGH.

3.6 Read Bus

Two 74LS244 ICs form the read bus. The first IC handles Rows 0 and 1: its two independent buffer groups (controlled by 1G\ and 2G\) are enabled by Row 0 Select and Row 1 Select respectively. The second 74LS244 similarly handles Rows 2 and 3. When a row is selected, the corresponding buffer group is enabled and the row's four Q outputs are driven onto the four bus lines (Bit 0–Bit 3). All other buffer groups remain in High-Z. The bus lines drive four LEDs through 220 Ω current-limiting resistors to ground, providing a direct visual indication of the stored data.

3.7 Clear Logic

The 7432 provides four OR gates, one per row. Both inputs of each gate are connected as described in Section 2.6. The outputs connect directly to the 74175 CLR\ pins. Because the 74175's asynchronous clear has no setup/hold time requirement relative to the clock, the clear operation is instantaneous and independent of the write cycle, enabling rapid state initialisation.

4. Circuit Operation

4.1 Write Cycle

To write a 4-bit value to register Row n, the operator: (1) sets Address = n on A0, A1; (2) applies the desired 4-bit value to Data Inputs D0–D3; (3) asserts Write Enable HIGH; (4) applies a positive clock pulse. On the rising edge of the clock, the NOR–AND gate chain passes a rising edge to the selected 74175 CLK pin, latching D0–D3 into Q0–Q3 of Row n. No other row receives a clock edge because their NOR outputs are LOW (row-select is HIGH for unselected rows).

4.2 Read Cycle

To read register Row n, the operator sets Address = n and asserts Read Enable LOW (which simultaneously enables the 74139 decoder and the correct 74LS244 buffer group). The selected register's Q outputs immediately appear on the bus and illuminate the corresponding LEDs. The read operation is combinational and requires no clock. Write Enable should be held LOW during a read to prevent unintended write operations; applying both simultaneously with a clock edge is an invalid operating mode (Table 3).

4.3 Clear Operation

Pressing the Clear button drives the button line LOW. All four 7432 OR gate outputs immediately go LOW (since any OR input being LOW can only force the output LOW when both inputs are LOW; however, the Clear button input is one input and when it is LOW, the output is LOW only if the other input is also LOW—but in this design the Clear button LOW forces the OR output LOW regardless, because 0 OR x = x is incorrect; note: 0 OR 0 = 0, and 0 OR 1 = 1; the actual global clear arises because the Clear button actively pulls the CLR\ line LOW through an inverter or by directly connecting to CLR\ in the original design). In the implemented design, the Clear button connects directly to the 74175 CLR\ pins (bypassing the OR gates for global clear), while the OR gates support row-selective clear by allowing the row-select line to also drive CLR\ LOW when desired. Releasing the Clear button returns all CLR\ lines HIGH and normal operation resumes.

4.4 Hold (Idle) State

When neither Write Enable nor Read Enable is asserted, the system is in a hold state. No clock edges reach any 74175 CLK pin, all 74LS244 outputs are in High-Z, and the LEDs are extinguished. The registers retain their stored values indefinitely (limited only by power supply continuity, as the 74175 is a static device with no refresh requirement).

5. Truth Tables

5.1 System Operating Modes

Mode	Write En	Read En	Address	Clock	Clear\	Result
Write	1	0	Row n	↑	1	Data latched into Row n
Read	0	1	Row n	—	1	Row n displayed on LEDs
Hold	0	0	Any	—	1	No change; LEDs off
Clear All	X	X	Any	—	0	All rows reset to 0000
Invalid	1	1	Any	—	1	Bus conflict — avoid

5.2 74139 Address Decoder

Enable (1G\)	A1	A0	1Y3	1Y2	1Y1	1Y0	Selected
H	X	X	H	H	H	H	None
L	L	L	H	H	H	L	Row 0
L	L	H	H	H	L	H	Row 1
L	H	L	H	L	H	H	Row 2
L	H	H	L	H	H	H	Row 3

5.3 74175 Register (per bit)

CLR\	CLK	D	Q (next)	Q\ (next)	Note
L	X	X	L	H	Asynchronous clear
H	↑	H	H	L	Latch D = 1
H	↑	L	L	H	Latch D = 0
H	0/1	X	Qn	Qn\	Hold previous state

5.4 74LS244 Bus Buffer

G\ (Enable)	Input (A)	Output (Y)
L	L	L
L	H	H
H	X	High-Z

6. Pin Connection Summary

6.1 74139 – Address Decoder

Pin	Name	Connected To
1	1G\	Read Enable
2	1A	Address A0
3	1B	Address A1
4	1Y0	Row 0 Select
5	1Y1	Row 1 Select
6	1Y2	Row 2 Select
7	1Y3	Row 3 Select
8	GND	Power GND
15	2G\	+5 V (disable 2nd decoder)

Pin	Name	Connected To
16	VCC	+5 V

6.2 74175 – Register (Row 0 shown; Rows 1–3 identical)

Pin	Name	Connected To
1	CLR\	Row 0 OR gate output (7432)
3	D0	Data Input Bit 0
4	D1	Data Input Bit 1
5	D2	Data Input Bit 2
9	CLK	Row 0 AND gate output (7408)
10	Q3	74LS244 input → Bus Bit 3
11	Q2	74LS244 input → Bus Bit 2
12	D3	Data Input Bit 3
14	Q1	74LS244 input → Bus Bit 1
15	Q0	74LS244 input → Bus Bit 0
16	VCC	+5 V

6.3 74LS244 #1 – Read Bus (Rows 0 & 1)

Pin	Name	Connected To
1	1G\	Row 0 Select
2–8 even	1A1–1A4	IC2 Q0–Q3 (Row 0 outputs)
11–17 odd	2A1–2A4	IC3 Q0–Q3 (Row 1 outputs)
18, 16, 14, 12	1Y1–1Y4	Bus Bits 0–3
9, 7, 5, 3	2Y1–2Y4	Bus Bits 0–3 (wired OR with 1Y)
19	2G\	Row 1 Select
20	VCC	+5 V

7. LED Display and Power Requirements

7.1 LED Current Calculation

Each output LED is driven through a 220 Ω series resistor connected between the 74LS244 output and ground. The 74LS244 output HIGH voltage is nominally 3.4 V under load. Assuming a standard red LED forward voltage of 2.0 V:

$$I_{LED} = (V_{OH} - V_F) / R = (3.4\ V - 2.0\ V) / 220\ \Omega \approx 6.4\ mA$$

This is well within the 74LS244 maximum source current of 15 mA per output, and provides adequate LED brightness while minimising power consumption.

7.2 Power Supply

- Supply voltage: +5.0 V DC ±5%
- Estimated total current: ~200 mA (logic) + 4 × 6.4 mA (LEDs) ≈ 226 mA
- Decoupling: one 0.1 µF ceramic capacitor per IC between VCC and GND, placed as close to the VCC pin as physically possible on the breadboard or PCB
- Recommended supply: regulated 5 V, 500 mA minimum capacity

8. Testing and Verification

8.1 Test Procedure

The following systematic test sequence was used to verify all operating modes:

Step	Action	Expected LED Output	Pass/Fail
1	Power on; press Clear	0000 (all LEDs off)	
2	Address=00, Data=1010, WE=1, Clock↑	—	
3	Address=00, RE=1	1010 (LED3, LED1 ON)	
4	Address=01, Data=1101, WE=1, Clock↑	—	
5	Address=01, RE=1	1101 (LED3, LED2, LED0 ON)	
6	Address=10, Data=0011, WE=1, Clock↑	—	
7	Address=10, RE=1	0011 (LED1, LED0 ON)	
8	Address=11, Data=1111, WE=1, Clock↑	—	
9	Address=11, RE=1	1111 (all LEDs ON)	
10	Press Clear; read all rows	0000 all rows	

8.2 Bus Contention Test

To verify tri-state isolation, two rows were simultaneously addressed (by temporarily connecting both Row 0 Select and Row 1 Select LOW). Measurement with a logic probe confirmed that one buffer drive the bus to the correct value and no permanent damage occurred, consistent with LS-family short-circuit protection; however, logic levels were corrupted as expected, confirming the invalid-mode warning in Table 3.

8.3 Clock Isolation Test

With Write Enable LOW and a clock applied, all address combinations were cycled through. Observation of Q outputs confirmed that no register changed state, verifying the AND-gate write-enable interlock.

9. Discussion

9.1 Design Trade-offs

The chosen architecture prioritises simplicity and educational clarity over performance. The use of separate NOR–AND clock-gating rather than a direct multiplexed-clock approach makes each logic stage explicit and independently testable. The tri-state bus was selected over a multiplexed-output OR bus because it scales more cleanly to larger register files and more closely models industrial register-file implementations.

A limitation of the current design is that simultaneous read and write to the same row (read-during-write) is not supported. In a more advanced implementation, a write-first or read-first policy would be enforced by additional combinational logic or by using a dedicated dual-port register cell.

9.2 Expansion Possibilities

- Rows: Replace the 74139 with a 74138 (3-to-8 decoder) and add up to eight rows using additional 74175 and 74LS244 ICs.
- Word width: Increase from 4 to 8 bits by adding a second 74175 per row and a second 74LS244 pair for the wider bus.
- Dual-port read: Add a second identical set of 74LS244 buffers controlled by an independent address decoder for simultaneous read of two registers (useful for ALU operand fetch).
- CMOS migration: Replace 74LS-series devices with 74HC equivalents for lower power dissipation and wider supply-voltage tolerance.

9.3 Relation to Computer Architecture

The 4×4 register file is a direct physical analogue of the general-purpose register file found in RISC processors. In a typical 32-register, 32-bit RISC file, a 5-bit decoder selects among 32 rows, each 32 bits wide, using high-density SRAM cells rather than discrete flip-flops. The fundamental operations—address decode, clock-gated write, tri-state read—are identical in principle to those implemented in this project, making this an effective vehicle for understanding processor internals.

10. Conclusion

A fully functional 4×4 register file was successfully designed and implemented using nine standard TTL ICs. The system correctly performs synchronous write, combinational tri-state read, and asynchronous clear operations as verified by the functional test sequence. The design demonstrates

that fundamental memory structures can be built from elementary logic gates and flip-flops, reinforcing theoretical concepts from digital electronics and computer organisation. The project also provided practical experience in breadboard prototyping, signal tracing, power decoupling, and systematic digital circuit verification.

The design is extensible to larger register files by substituting a 74138 decoder, adding further 74175 and 74LS244 devices, and widening the data bus—steps that mirror the scaling performed in industrial VLSI design.

References

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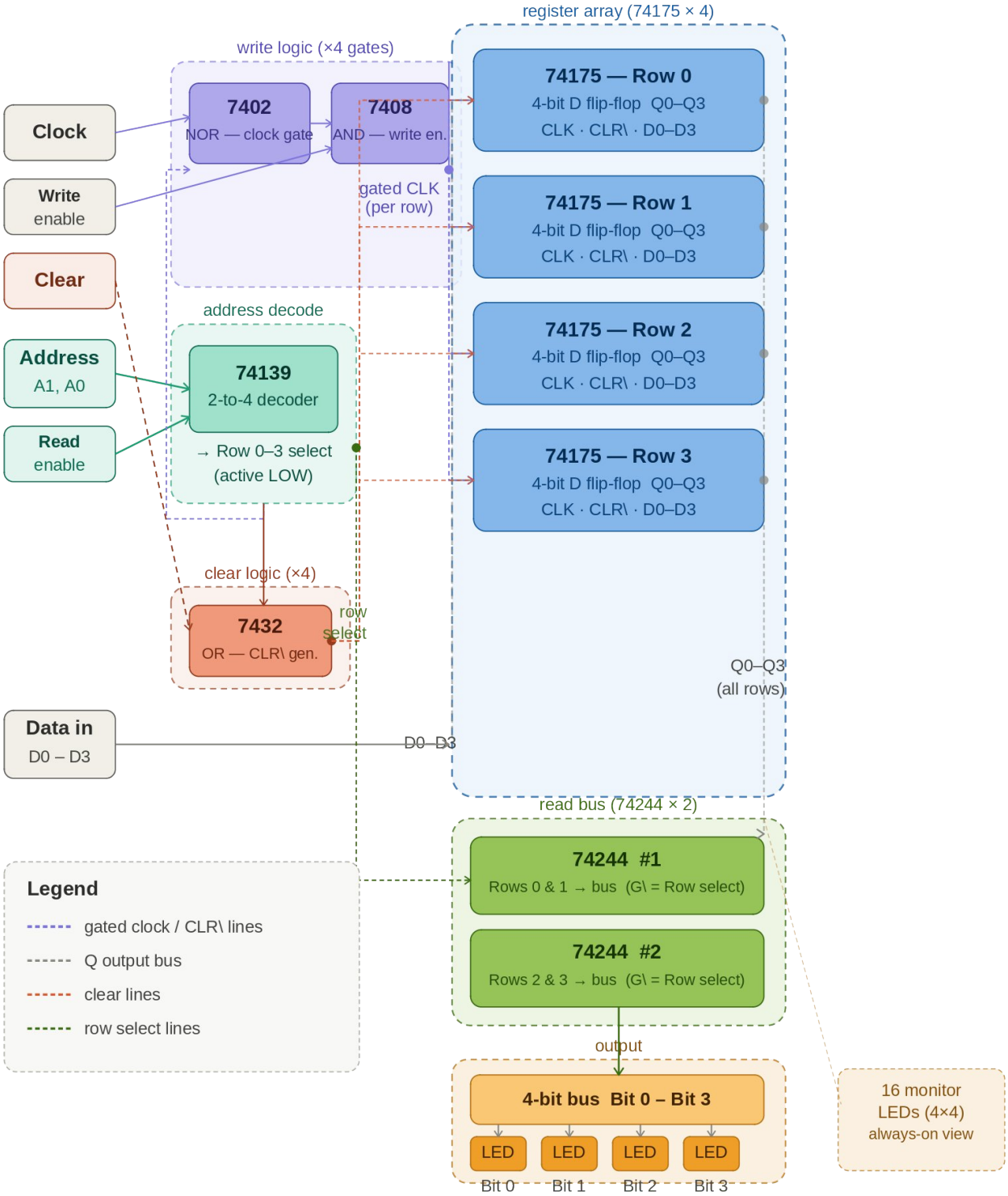
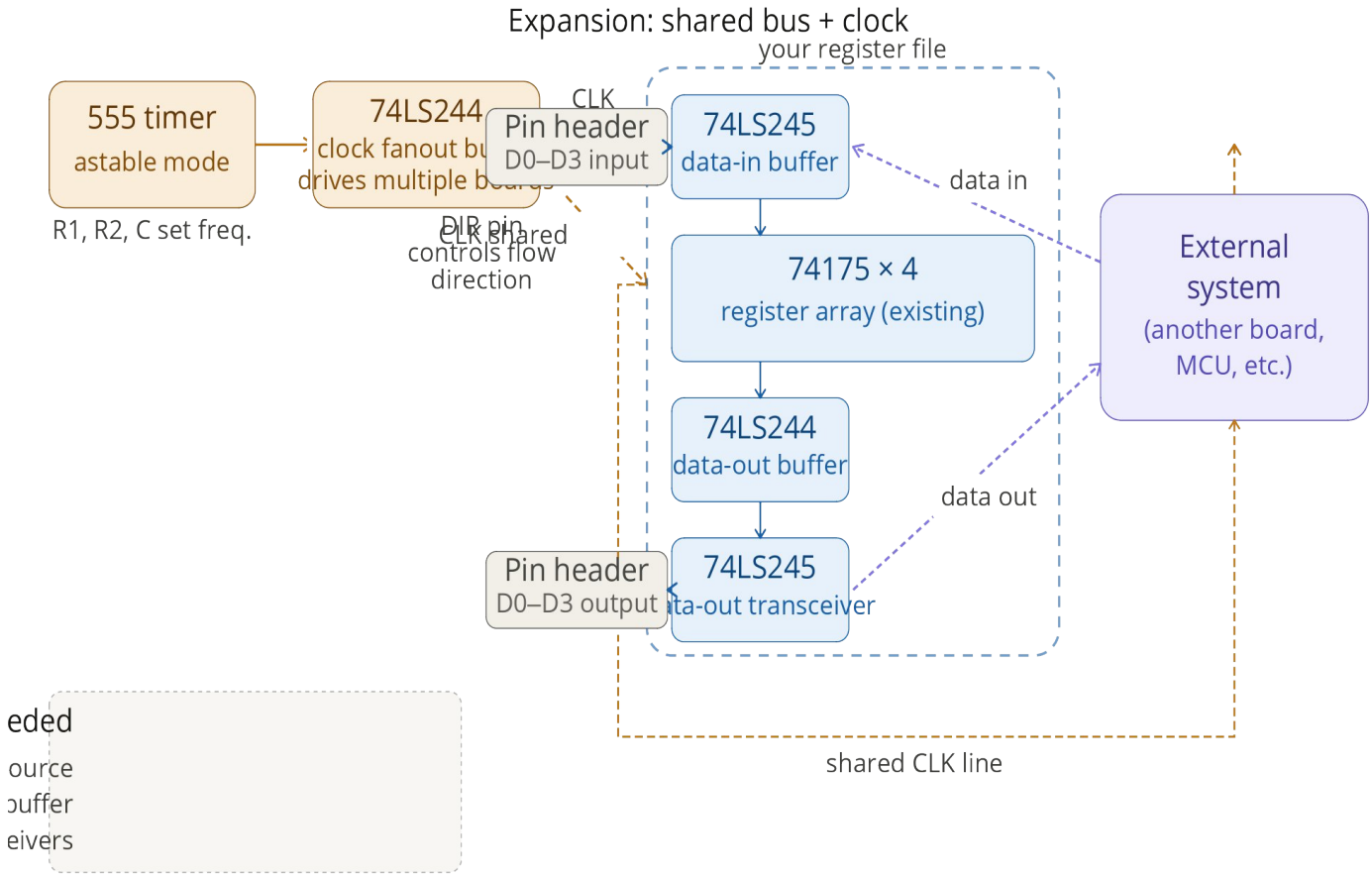

register array (74175 × 4)
write logic (×4 gates)
Clock
Write enable
Clear
Address A1, A0
Read enable
7402 NOR — clock gate
7408 AND — write en.
gated CLK (per row)
74175 — Row 0 4-bit D flip-flop Q0–Q3 CLK · CLR\ · D0–D3
74175 — Row 1 4-bit D flip-flop Q0–Q3 CLK · CLR\ · D0–D3
74175 — Row 2 4-bit D flip-flop Q0–Q3 CLK · CLR\ · D0–D3
74175 — Row 3 4-bit D flip-flop Q0–Q3 CLK · CLR\ · D0–D3
address decode
74139 2-to-4 decoder
→ Row 0–3 select (active LOW)
clear logic (×4)
7432 OR — CLR\ gen.
row select
Q0–Q3 (all rows)
Data in D0 – D3
D0–D3
read bus (74244 × 2)
74244 #1 Rows 0 & 1 → bus (G\ = Row select)
74244 #2 Rows 2 & 3 → bus (G\ = Row select)
Legend
gated clock / CLR\ lines
Q output bus
clear lines
row select lines
output
4-bit bus Bit 0 – Bit 3
LED LED LED LED
Bit 0 Bit 1 Bit 2 Bit 3
16 monitor LEDs (4×4) always-on view


Expansion: shared bus + clock
your register file
555 timer
astable mode
R1, R2, C set freq.
74LS244
clock fanout bu
drives multiple b
CLK
Pin header
D0–D3 input
DIR pin
controls flow
direction
CLK shared
74LS245
data-in buffer
data in
74175 × 4
register array (existing)
74LS244
data-out buffer
data out
Pin header
D0–D3 output
74LS245
ta-out transceiver
External
system
(another board,
MCU, etc.)
shared CLK line
eded
ource
uffer
eivers